

Psychopathology

Depression Detectable In Voice: AIIMS Researchers

At an advanced Speech Health Lab set up at AIIMS Delhi with CSR support, researchers analysed speech samples from 423 participants with complete clinical and demographic records. The mean participant age was about 24 years, with most between 18 and 25.



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Published On Feb 2, 2026 at 03:10 PM IST



New Delhi: The way a person speaks — from tone and fluency to emotional colouring and vocal effort — may offer early clues to depression, according to research underway at the All India Institute of Medical

Sciences (AIIMS) Delhi.

Some people overreact to simple statements and situations. Some have difficulty in relating to other people, some often find fault with their colleagues and family members and feel threatened; others seem to experience hallucinations and delusions. Often we dismiss these people sometimes casually and sometimes with fear, as “mad” or “eccentric.”

- Morbidity on account of mental illness is set to overtake cardiovascular diseases as the single largest risk in India.
- According to NIMHANS, there are over two crore people in India who are in need of treatment for serious mental disorders and about five crore people who are affected by common mental disorder.
- About 30 to 35 lakh people need hospitalization at any time for mental illness. As Mohandas (2009) has observed the situation is alarming.
- He noted that “a meta-analysis of 13 epidemiological studies consisting of 33,572 persons reported a total morbidity of 58.2 per 1000. Another meta-analysis of 15 epidemiological studies reported a total morbidity of 73 per 1000.
- The number of psychiatric beds in the country is only about 0.2 per 1, 00,000 population and there are only two psychiatrists per 10 lakh population

- Math and Srinivasraju (2010) have estimated that about 20 percent of the adult population in the community is affected with one or the other psychiatric disorder.
- In particular female gender, children and adolescents, students, aged people, people suffering from chronic medical condition, disabled, disaster survivors, people in custodial care, marginalized persons, refugees and individuals with poor family, social and economic support are found at high risk of develop psychiatric disorders.

Mental Disorder

Most psychologists do agree that mental disorders include the following features.

First, they involve patterns of behavior or thought that are judged to be unusual or atypical in the society. People with these disorders don't behave or think like most others, and these differences are often apparent to the people around them.

Second, such disorders usually generate distress— negative feelings and reactions—in the persons who experience them.

Third, mental disorders are maladaptive—they interfere with individuals' ability to function normally and meet the demands of daily life.

Mental disorders as disturbances of an individual's behavioral or psychological functioning that are not culturally accepted and that lead to psychological distress, behavioral disability, and/or impaired overall functioning (Nietzel et al., 1998).

Pervasiveness (in psychopathology) means that a pattern of thoughts, emotions, or behaviors occurs across many situations and life domains—not just in one specific context.

Persistence in psychopathology refers to how long symptoms last and whether they continue over time, even when the original trigger or situation has changed.

1. Is there significant distress?
2. Is there functional impairment? (academics, relationships, self-care)
3. Is it persistent and inflexible? (not just situational or short-term)
4. Is it culturally/contextually inappropriate?

A student feels anxious before an important presentation, but the anxiety reduces once the presentation is over and does not affect attendance or performance.

Is this psychopathology?

- A. Yes, because anxiety is present
- B. No, because the anxiety is situational and temporary
- C. Yes, because anxiety always indicates a disorder
- D. Cannot be determined

Which of the following is the strongest indicator that a behavior may be psychopathological?

- A. The behavior is unusual
- B. The behavior causes significant functional impairment
- C. The behavior is emotionally intense
- D. The behavior is disliked by others

A pattern of emotional instability and interpersonal conflict has been present since late adolescence and affects work, friendships, and self-image.

This example best illustrates which feature of psychopathology?

- A. Situational reaction
- B. Temporary distress
- C. Pervasiveness and persistence
- D. High emotionality

Which combination best helps distinguish psychopathology from normal distress?

- A. Intensity + uniqueness
- B. Duration + cultural acceptability
- C. Distress + impairment + persistence
- D. Emotional pain + stressor

Models of Abnormality

Frameworks that provide comprehensive accounts of how and why mental disorders develop and how they can best be treated.

Diathesis-Stress Model

A model suggesting that genetic or other factors may predispose an individual to develop a mental disorder by that the disorder will develop only if the person is exposed to certain kinds of stressful environmental conditions.

This view suggests that mental disorders result from the joint effects of two influences:

- (1) a predisposition for a given disorder, termed a diathesis, and
- (2) stressors in an individual's environment that tend to activate or stimulate the predisposition or vulnerability

In other words, the diathesis–stress model suggests that for various reasons—genetic factors, early traumatic experiences, specific personality traits—individuals show varying degrees of vulnerability to specific mental disorders.

Development Psychopathology Perspective: A view emphasizing the fact that problems that first appear during childhood or adolescence are often precursors for disorders that occurs later in life.

Several different systems for classifying mental disorders exist (e.g., the International Statistical Classification of Diseases, Injuries, and Causes of Death, or ICD-10 for short; World Health Organization, 1992).

However, the one that is the most widely used in the United States is the Diagnostic and Statistical Manual of Mental Disorders–IV (DSM–IV), published by the American Psychiatric Association (1994).

How psychologists assess?

1. Life Records
2. Assessment Interviews
3. Psychological Tests
4. Observations of Behavior
5. Biological Measures

Mood Disorders

Have you ever felt truly “down in the dumps”—sad, blue, and dejected? How about “up in the clouds”—happy, elated, excited? Probably you can easily bring such experiences to mind, for everyone has swings in mood or emotional state. For most of us, these swings are usually moderate in scope; periods of deep despair and wild elation are rare. Some persons, however, experience swings in their emotional states that are much more extreme and prolonged. Their highs are higher, their low than most people. Such persons are described as suffering from mood disorders. Among the most important of these are depressive disorders and bipolar disorders.

Depression: A mood disorder in which individuals experience extreme unhappiness, lack of energy, and several related symptoms.

Symptoms of depression

A. Five (or more) of the following symptoms have been present during the same two-week period and represent a change from previous functioning; at least one of the symptoms is either (1) depressed mood or (2) loss of interest or pleasure.

NOTE: Do not include symptoms that are clearly attributable to another medical condition.

- 1) Depressed mood most of the day, nearly every day, as indicated by either subjective report (eg, feels sad, empty, hopeless) or observations made by others (eg, appears tearful). (NOTE: In children and adolescents, can be irritable mood.)
- 2) Markedly diminished interest or pleasure in all, or almost all, activities most of the day, nearly every day (as indicated by either subjective account or observation).
- 3) Significant weight loss when not dieting or weight gain (eg, a change of more than 5% of body weight in a month), or decrease or increase in appetite nearly every day. (NOTE: In children, consider failure to make expected weight gain.)
- 4) Insomnia or hypersomnia nearly every day.

- 5) Psychomotor agitation or retardation nearly every day (observable by others, not merely subjective feelings of restlessness or being slowed down).
 - 6) Fatigue or loss of energy nearly every day.
 - 7) Feelings of worthlessness or excessive or inappropriate guilt (which may be delusional) nearly every day (not merely self-reproach or guilt about being sick).
 - 8) Diminished ability to think or concentrate, or indecisiveness, nearly every day (either by their subjective account or as observed by others).
 - 9) Recurrent thoughts of death (not just fear of dying), recurrent suicidal ideation without a specific plan, or a suicide attempt or a specific plan for committing suicide.
- B.** The symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- C.** The episode is not attributable to the direct physiological effects of a substance or to another medical condition.



Causes of Depression

1. **Biological Underpinnings:**

- Depression tends to run in families and is about four times more likely to occur in both members of identical-twin pairs than in both members of nonidentical-twin pairs (Bowman & Nernberger, 1993).
- It has been found that levels of two neurotransmitters, norepinephrine and serotonin, are lower in the brains of depressed persons than in those of nondepressed persons.
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- Hormonal level.

2. **learned helplessness** (Seligman, 1975), or beliefs on the part of individuals that they have no control over their own outcomes. Such views often develop after exposure to situations in which such lack of control is present, but then generalize to other situations where individuals' fate is at least partly in their hands. One result of such feelings of helplessness seems to be depression (e.g., Seligman et al., 1988).

3. **negative self-schemas**—negative conceptions of their own traits, abilities, and behavior. As a result, they tend to be highly sensitive to criticism from others (Joiner, Alfano, & Metalsky, 1993). Because such persons are more likely to notice and remember negative information, their feelings of worthlessness strengthen; and when they are exposed to various stressors (e.g., the breakup of a romantic relationship, a failure at work), their thinking can become distorted in important and self-defeating ways.



Explanatory styles

- *The defining characteristic of pessimists is that they tend to believe that bad events will last a long time, will undermine everything they do, and are their own fault.*
- optimistic explanatory style assume that situations will work out for the best in the end.
- 3 dimensions based on personalization, permanence, and pervasiveness.



Dimensions of explanatory styles

Internal An individual with a propensity to blame failure/success on themselves Positive internalization=optimism Negative internalization=pessimism	External An individual with a propensity to blame failure/success on something external Positive externalization=pessimism Negative externalization=optimism
Stable (Weiner, 1972) Considers any event as permanent Positive stable=optimism Negative stable=pessimism	Unstable Considers any life event as temporary Positive unstable=pessimism Negative unstable=optimism



Global

A global attribution occurs when an individual attributes an outcome to a factor they perceive to be consistent, irrespective of context.

Positive global=optimism

Negative global=pessimism

Specific

A specific attribution occurs when an individual attributes an outcome to a factor only relevant in the specific context or setting of the experience.

Positive specific=pessimism

Negative specific=optimism

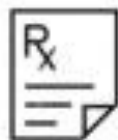


	Good Situation	Bad Situation
Optimist	Permanent Pervasive Personal (Internal)	Temporary Specific External cause
Pessimist	Temporary Specific External cause	Permanent Pervasive Personal (internal)

Major causes of depression



Brain chemistry imbalance



Physical health problems



Stress factors



Genetics and biology



Drugs



Traumatic event



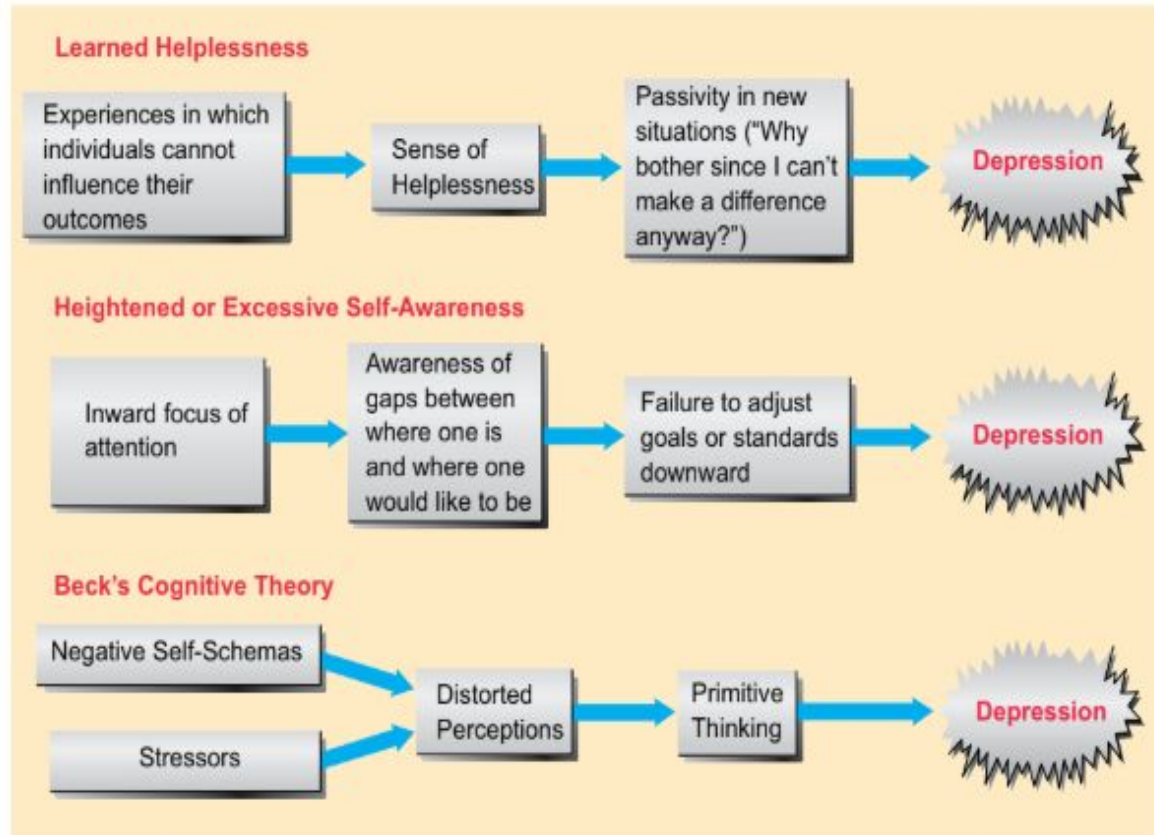
Nutritional disorder



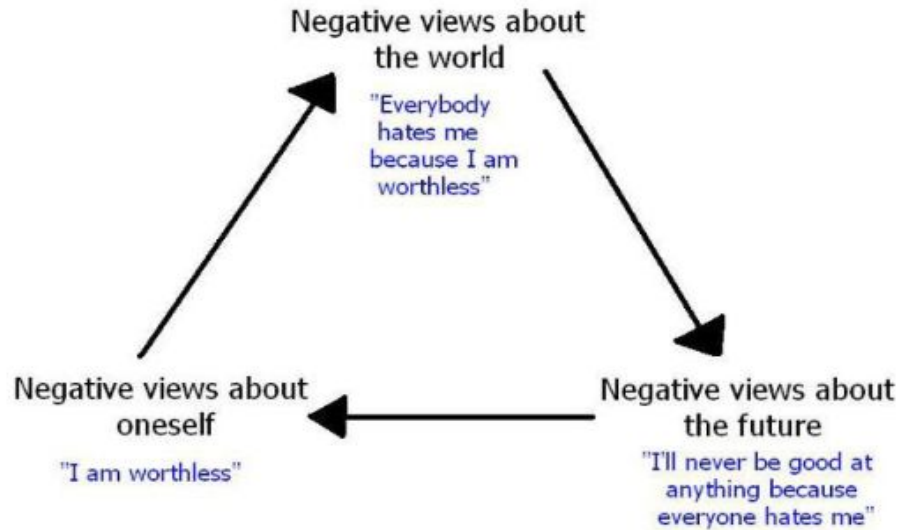
Female sex hormones

Cognitive Mechanisms in Depression

Extensive research indicates that the cognitive mechanisms and factors shown here often play a role in depression.



Beck's cognitive triad





At the root are **maladaptive schemas**—deep-seated core beliefs developed early in life often through,

- loss
- criticism
- rejection
- failure experiences



Bipolar Disorder

- Emotional roller coaster
- Mood swings between depression and mania/hypomania
- Most genetically influenced of all mental disorders
- Higher risk of suicide than are most other disorders



Symptoms of a manic episode

- Having an abnormally high level of activity or energy.
- Feeling extremely happy or excited — even euphoric.
- Not sleeping or only getting a few hours of sleep but still feeling rested.
- Having inflated self-esteem, thinking you're invincible.
- Being more talkative than usual. Talking so much and so fast that others can't interrupt.
- Having racing thoughts — having lots of thoughts on lots of topics at the same time (called a “flight of ideas”).
- Being easily distracted by unimportant or unrelated things.
- Being obsessed with and completely absorbed in an activity.
- Displaying purposeless movements, such as pacing around your home or office or fidgeting when you're sitting.
- Showing impulsive behavior that can lead to poor choices, such as buying sprees, reckless sex or foolish business investments.

Bipolar 1

Bipolar 2



Mania

At least one episode of extreme mania lasting more than a week.

Symptoms of hypomania (a milder form of mania) lasting at least 4 days.



Depression

Tends to be milder than other bipolar types. Some experience no depression.

At least one depressive episode, broken up by periods of hypomania.



Symptoms

- Increased energy
- Talking extremely quickly
- Euphoria

- Feeling of hopelessness
- Fatigue
- Irritable and anxious

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- Anxiety: Increased arousal accompanied by generalized feelings of fear or apprehension.
 - State Anxiety: State anxiety refers to a temporary emotional condition characterized by feelings of tension, nervousness, worry, and physiological arousal in response to a specific situation.
 - Trait Anxiety: Trait anxiety is a stable personality characteristic that reflects a person's general tendency to respond with anxiety across many situations.
 - High trait anxiety → More frequent/intense state anxiety



Generalised Anxiety Disorder:

Persistent and excessive worry that tends to interfere with daily activities.



Phobias:

Persistent and excessive fear around a particular object, activity, or situation.



Social Anxiety Disorder:

Intense anxiety about being embarrassed or rejected in social situations.



Panic Disorder:

The main symptom is panic attacks, physical and psychological distress episodes.





When assessing for GAD, clinical professionals are looking for the following:

The presence of excessive anxiety and worry about a variety of topics, events, or activities. Worry occurs more often than not for at least six months and is clearly excessive.

- The worry is experienced as very challenging to control. The worry in both adults and children may easily shift from one topic to another.
- The anxiety and worry are accompanied by at least three of the following physical or cognitive symptoms (In children, only one of these symptoms is necessary for a diagnosis of GAD):
 - Edginess or restlessness
 - Tiring easily; more fatigued than usual
 - Impaired concentration or feeling as though the mind goes blank
 - Irritability (which may or may not be observable to others)
 - Increased muscle aches or soreness
 - Difficulty sleeping (due to trouble falling asleep or staying asleep, restlessness at night, or unsatisfying sleep)



Why do people worry

1. Individuals with GAD demonstrate cognitive and affective hyperreactivity
2. Worry as an experiential avoidance mechanism

a current notion in the GAD literature is that worry may be a cognitive mechanism employed in an attempt to master problems with emotional dysregulation and to feel more in control. To this end, Borkovec and colleagues' **Cognitive Avoidance theory** of worry (Borkovec et al. 2004) has played an important role in helping to inform our understanding of the relationship between worry and emotional dysfunction.

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1. Worry leads to greater concurrent negative emotionality and physiological activation
 2. Worry does not suppress subsequent negative emotionality or physiological activation
 3. Worry prolongs negative emotionality and physiological activation
 4. Worry as a problematic compensatory strategy



Risk Factors

1. Environmental. The development of GAD is associated with unexpected **negative life events (Nordahl et al. 2010), maltreatment (Moffitt et al. 2007), and loss (Kendler et al. 2003)**. Such adverse events can lead people to question their worldviews—that life is predictable and good things happen to good people. Even the presence of a one-time negative life event may lead individuals to become continuously anxious as a means to prepare for other potentially unpredictable events.



2. Attachment and parenting style. **Insecure attachment and dysregulation of emotions**, such as anxiety, go hand in hand as potentially important developmental risk factors.

Negative parenting behaviors have also been associated with GAD symptoms and worry. In community samples, samples of anxious children, and clinical samples of adults, **perceived parental rejection** was associated with offspring worry (Brown & Whiteside 2008, Muris et al. 2000b) and GAD symptoms (Cassidy et al. 2009, Hale et al. 2006). Similarly, retrospective reports of **maternal and paternal coldness** (Cassidy et al. 2009) were associated with adult GAD.



According to DSM-5, a panic attack is characterized by four or more of the following symptoms (the presence of fewer than four symptoms may be considered a limited-symptom panic attack):

- Palpitations, pounding heart, or accelerated heart rate
- Sweating
- Trembling or shaking
- Sensations of shortness of breath or smothering
- A feeling of choking
- Chest pain or discomfort
- Nausea or abdominal distress
- Feeling dizzy, unsteady, lightheaded, or faint
- Feelings of unreality (derealization) or being detached from oneself (depersonalization)
- Fear of losing control or going crazy
- Fear of dying
- Numbness or tingling sensations (paresthesias)
- Chills or hot flushes

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- Condition characterized by periodic, unexpected attacks of intense, terrifying anxiety known as panic attacks.
 - repeated and unexpected panic attacks, along with either persistent concerns about future attacks or a change in personal behavior in an attempt to avoid triggers
 - episodic and sudden, not constant worry

Causes Of Anxiety Attacks & Panic Attacks



**Stressful
environments**



**Drug or alcohol
withdrawal**



**Chronic
conditions**



Medications



**Exposure to
trauma triggers**



**Overuse of
stimulants,
including caffeine**





Unexpected Panic attacks:

An unexpected panic attack occurs without an obvious trigger. It seems to come “out of the blue.”

Expected Panic attacks:

An expected panic attack occurs in response to a specific feared situation or trigger.

Cognitive Model of Panic Attacks

